Binom

```
count <- c(3,7)
n <- sum(count)
chi2 <- chisq.test(count, p=c(0.5,0.5), correct = FALSE)
chi2
```

```
##
##  Chi-squared test for given probabilities
##
## data:  count
## X-squared = 1.6, df = 1, p-value = 0.2059
```

```
t = sqrt(n)*(count[1]/n-0.5)/0.5
t^2
```

```
## [1] 1.6
```

```
pnorm(t)
```

```
## [1] 0.1029516
```

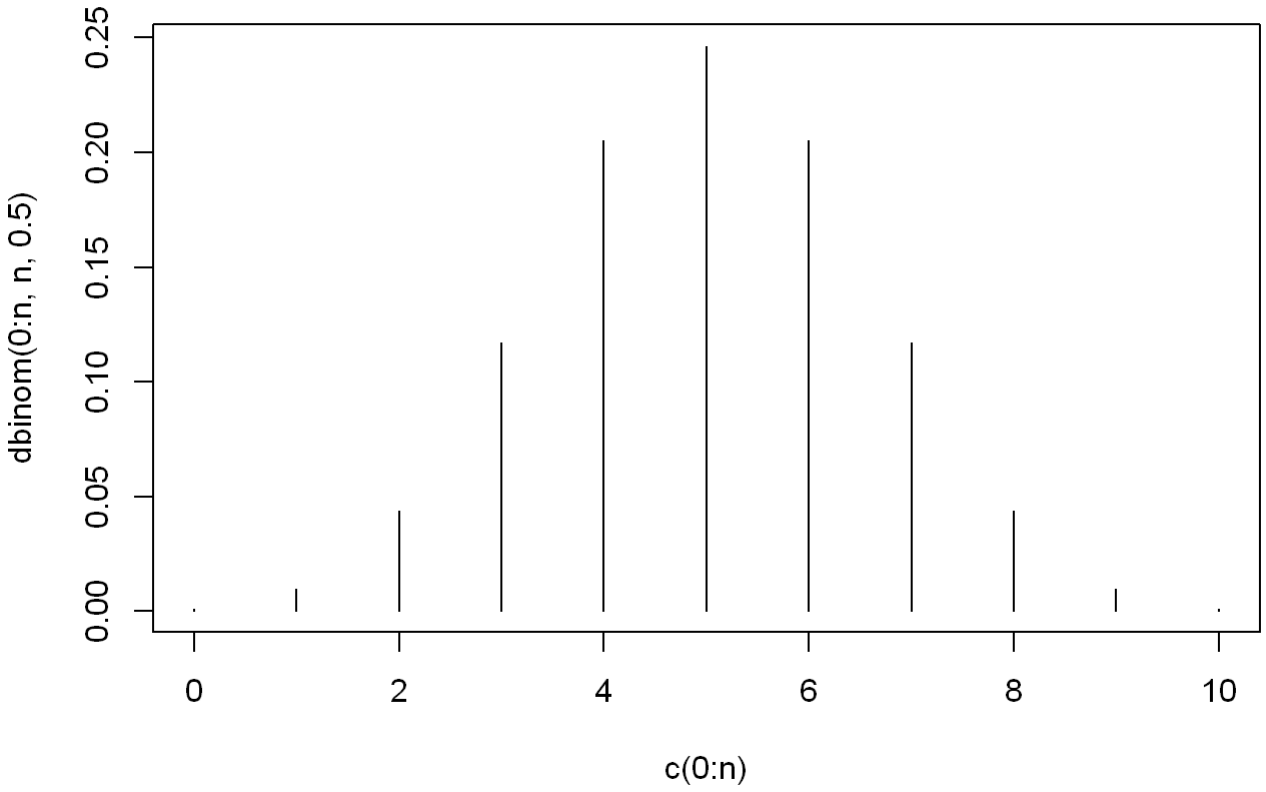
```
2*pnorm(t)
```

```
## [1] 0.2059032
```

```
df <- data.frame(0:n, dbinom(0:n, n, 0.5), pbinom(0:n, n, 0.5), 1-pbinom(0:n, n, 0.5))
df
```

```
##    X0.n dbinom.0.n..n..0.5. pbinom.0.n..n..0.5. X1...pbinom.0.n..n..0.5.
## 1     0        0.0009765625        0.0009765625             0.9990234375
## 2     1        0.0097656250        0.0107421875             0.9892578125
## 3     2        0.0439453125        0.0546875000             0.9453125000
## 4     3        0.1171875000        0.1718750000             0.8281250000
## 5     4        0.2050781250        0.3769531250             0.6230468750
## 6     5        0.2460937500        0.6230468750             0.3769531250
## 7     6        0.2050781250        0.8281250000             0.1718750000
## 8     7        0.1171875000        0.9453125000             0.0546875000
## 9     8        0.0439453125        0.9892578125             0.0107421875
## 10    9        0.0097656250        0.9990234375             0.0009765625
## 11   10        0.0009765625        1.0000000000             0.0000000000
```

```
plot(dbinom(0:n, n, 0.5) ~ c(0:n), type = "h")
```




```
bt <- binom.test(count[1], n, p = 0.5,
           alternative = "less",
           conf.level = 0.95)
bt
```

```
##
##  Exact binomial test
##
## data:  count[1] and n
## number of successes = 3, number of trials = 10, p-value = 0.1719
## alternative hypothesis: true probability of success is less than 0.5
## 95 percent confidence interval:
##  0.0000000 0.6066242
## sample estimates:
## probability of success
##                    0.3
```